$$x_n = \int_1^n \frac{\cos t}{t^2}\, dt$$

show: <u>Cauchy sequence.</u>

Hint: $\displaystyle\int_1^n \left|\frac{\cos t}{t^2}\right| dt \le \int_1^n \frac{1}{t^2}\, dt$

$$\boxed{d(x_n, x_m) \xrightarrow[n,m \to \infty]{} 0}$$

Try it: $d(x_n, x_m) = |x_n - x_m|$

$$= \left| \int_1^n \frac{\cos t}{t^2}\, dt - \int_1^m \frac{\cos t}{t^2} \right|$$

Assume $n > m$ $\quad = \left| \displaystyle\int_m^n \frac{\cos t}{t^2}\, dt \right|$

Triangle inequality. $\quad \le \displaystyle\int_m^n \left| \frac{\cos t}{t^2} \right| dt$

Aside:

$|-3+2| \le |-3| + |2|$

Triangle ineq.

$d(x,y)$: metric

$$d_b(x,y) = \frac{d(x,y)}{1 + d(x,y)}$$

To show: $d_b(x,y) \le d_b(x,z) + d_b(y,z)$

To show: $\dfrac{d(x,y)}{1+d(x,y)} \le \dfrac{d(x,z)}{1+d(x,z)} + \dfrac{d(y,z)}{1+d(y,z)}$.

Let $a = d(x,y)$
$b = d(x,z)$
$c = d(y,z)$

We know

$a \le b + c$

To show: $\dfrac{a}{1+a} \overset{?}{\le} \dfrac{b}{1+b} + \dfrac{c}{1+c}$

To show:

$$a(1+b)(1+c) \overset{?}{\le} b(1+a)(1+c) + c(1+a)(1+b)$$

$$a \le b + c$$

$$\left\| \begin{bmatrix} x \\ y \end{bmatrix} \right\|_p = \left[|x|^p + |y|^p \right]^{1/p}$$

If $|x| > |y|$:

$$|x| - |y| > 0$$

$$\frac{|y|}{|x|} < 1$$

$$\left[|x|^p \left(1 + \left(\frac{|y|}{|x|} \right)^p \right) \right]^{1/p}$$

$$|x| \left(1 + \left| \frac{y}{x} \right|^p \right)^{1/p}$$

$$\approx |x| \left(1 + \left| \frac{y}{x} \right|^p \frac{1}{p} \right)$$

$$\left[|x|^p |y|^p \right]^{1/p}$$

$$= |x| \cdot |y|$$

Aside

$$(a+b)^n = \underbrace{a^n + \binom{n}{1} a^{n-1} b}_{} + \cdots + \binom{n}{n-1} a b^{n-1} + b^n$$

$$\boxed{\begin{array}{l} b \ll 1 \\ (a+b)^n \approx a^n + nb \end{array}}$$